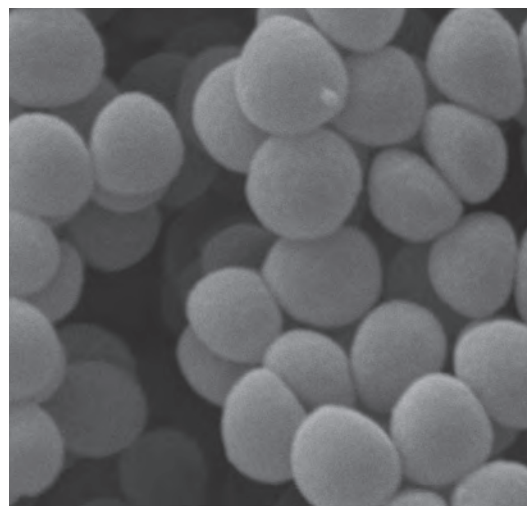
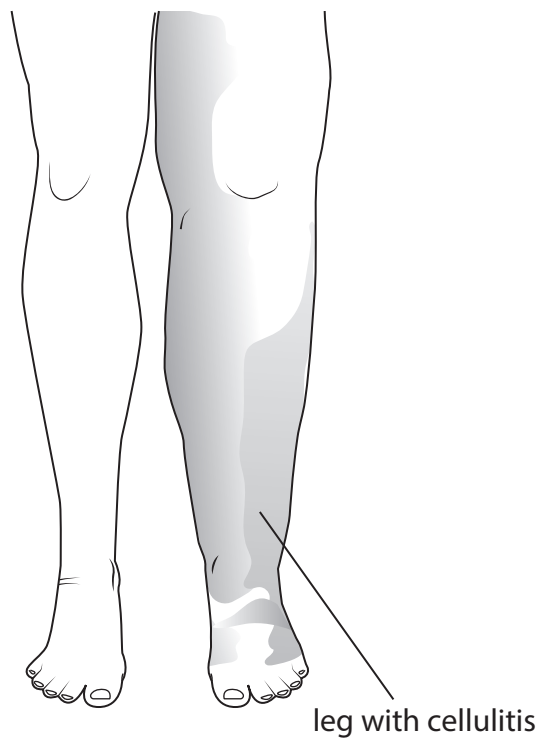


7 Cellulitis is an infection of the skin.

It is caused when bacteria living on the skin surface enter through a cut in the skin.

This results in swelling and redness of the infected area.

The images show a patient with cellulitis in one leg and one of the types of skin bacteria that cause cellulitis.



(Source: © Louis Lau / Alamy Stock Photo)

bacteria

(a) This type of bacteria has a capsule, but not a flagellum.

(i) Suggest **one** advantage of having a capsule to this type of bacteria.

(1)

(ii) Suggest why this type of bacteria does **not** need a flagellum.

(1)

(iii) Which of the following structures would be found in a bacterial cell?

1. cell membrane
2. cell wall
3. ribosomes

(1)

- ☐ **A** 1 and 2 only
- ☐ **B** 1 and 3 only
- ☐ **C** 2 and 3 only
- ☐ **D** 1, 2 and 3

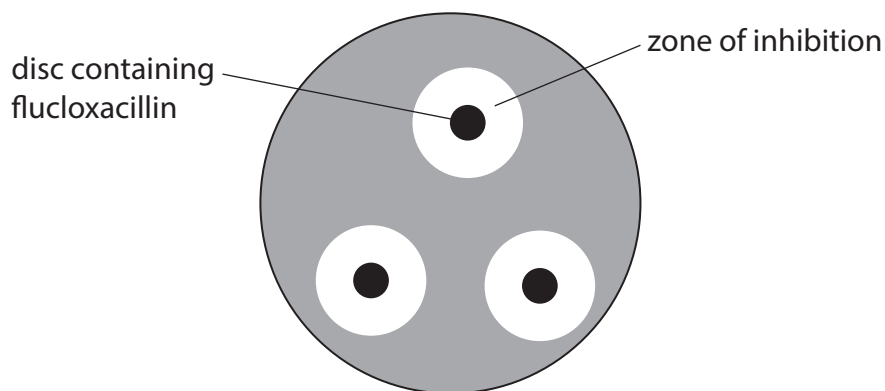
(b) The antibiotic flucloxacillin is the current treatment for cellulitis.

A student investigated the antimicrobial properties of flucloxacillin.

An agar plate had the type of bacteria which causes cellulitis spread over the surface of the agar.

Three discs containing the **same** concentration of flucloxacillin were placed onto the plate, as shown in the diagram.

The plates were incubated for 24 hours at 25°C.



The table shows the student's results.

Diameter of zone of inhibition / mm
11.0
7.5
8.5
Mean ($\bar{x}$) = 9.0



Standard deviation can be calculated by using the formula:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

The table shows some of the calculations done by the student.

x	$x - \bar{x}$	$(x - \bar{x})^2$
11.0	2.0	4.0
7.5		
8.5	-0.5	0.25
		$\sum (x - \bar{x})^2 =$

- (i) Calculate the standard deviation for this set of data.

Use the student's table and formula to help you.

(3)

Answer mm

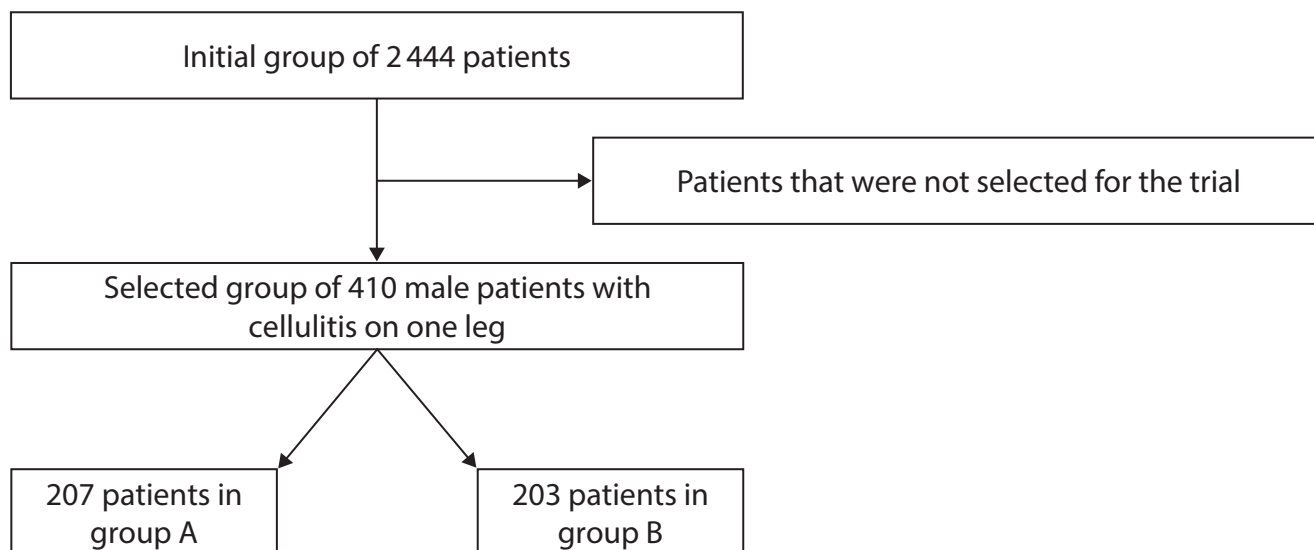
- (ii) Which of the following explains why the student would want to calculate the standard deviation(s) (SD) for this experiment?

(1)

- ☐ **A** to see how widely the diameter values differ from the mean diameter
- ☐ **B** to see if the observed results differ from the expected results
- ☐ **C** to see if there is a correlation between different concentrations and zone of inhibition diameter
- ☐ **D** to see which is the most frequently occurring number

- (c) The effectiveness of treating cellulitis with flucloxacillin alone was compared with a treatment that used both flucloxacillin and clindamycin.

The diagram shows how two groups of patients with cellulitis were selected from an initial group of 2 444 patients.



- (i) Which percentage of the initial group of patients were placed in **group B**?

(1)

- ☐ **A** 8.3 %
- ☐ **B** 8.5 %
- ☐ **C** 49.5 %
- ☐ **D** 50.5 %

- (ii) Suggest how this study could be carried out as a double-blind trial.

Use the diagram to support your answer.

(1)

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.....

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- (iii) The table shows the mean difference in circumference for the infected and non-infected legs at 5 and 10 days after treatment began.

Group	Mean difference in circumference between infected leg and non-infected leg / cm	
	After 5 days	After 10 days
A	2.18	1.77
B	2.04	1.42

Comment on the results of this investigation.

(4)

(Total for Question 7 = 13 marks)

